Interactive Diagram Demo: Recursion

Published: March 12, 2026

Reading the sequence

The growing activation bars represent pending stack frames; the dashed arrows show values returning as the stack unwinds.

The submission buttons above are placeholders. They open `example.com` and do not submit data.

Evaluating factorial(3)

Each call waits for the next smaller factorial before it can return.

Call factorial(3)

Main starts the recursive calculation.

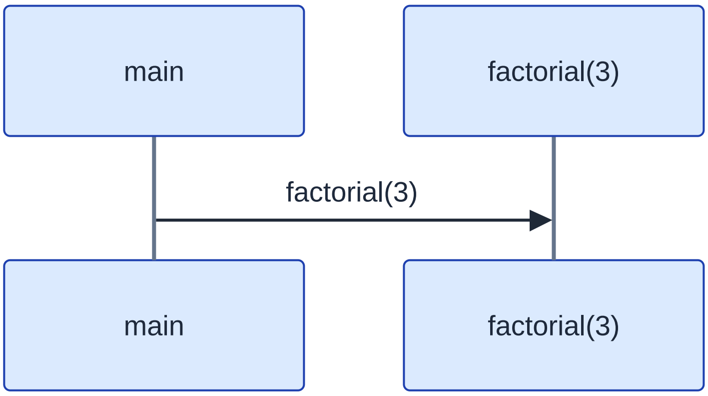

Figure 1: Call factorial(3)

Recurse to factorial(2)

factorial(3) needs the value of factorial(2).

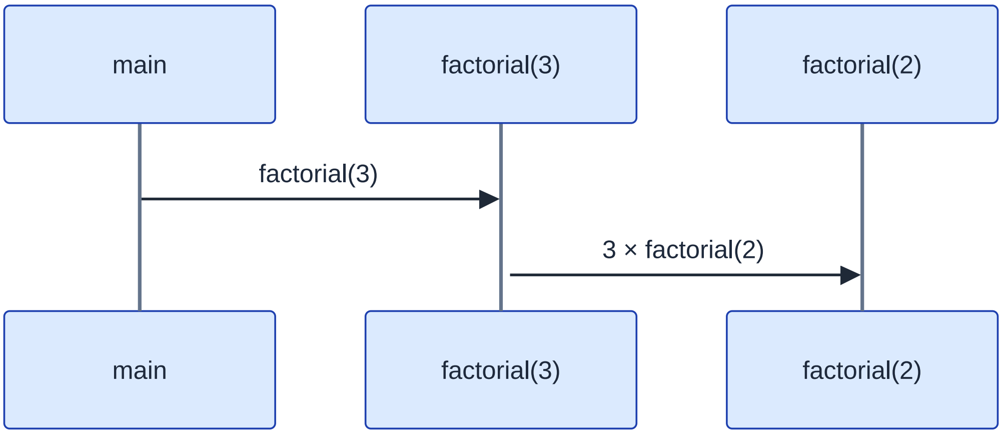

Figure 2: Recurse to factorial(2)

Reach the base case

factorial(2) calls factorial(1), which returns 1.

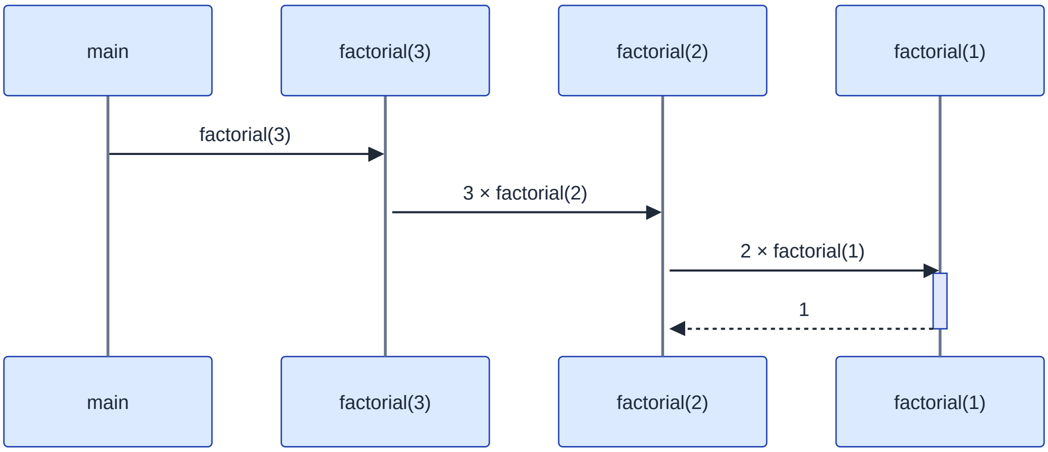

Figure 3: Reach the base case

Unwind factorial(2)

factorial(2) multiplies 2 by 1 and returns 2.

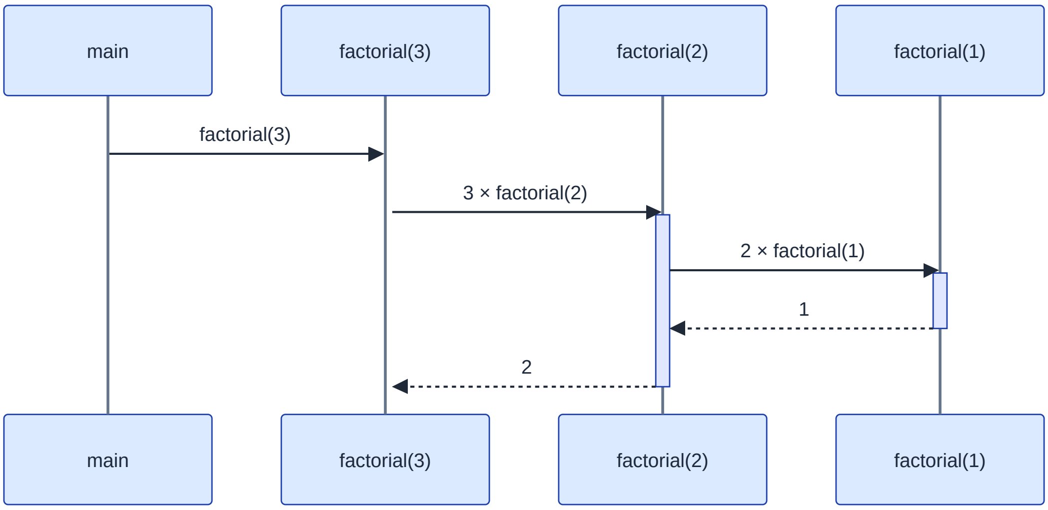

Figure 4: Unwind factorial(2)

Return the final result

factorial(3) multiplies 3 by 2 and returns 6 to main.

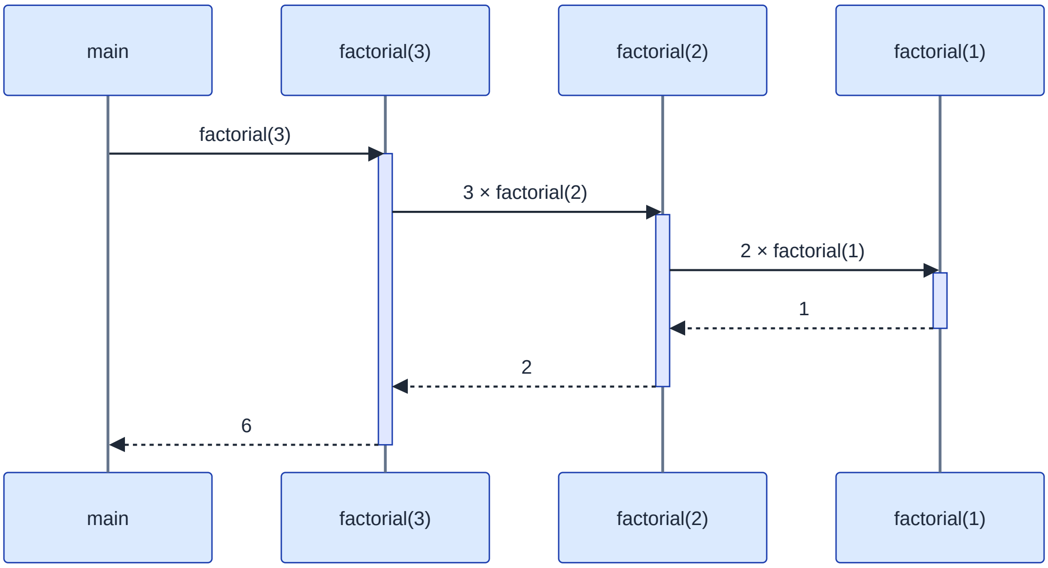

Figure 5: Return the final result